Compliance Laboratory

Test Report

Applicant:	Yongkang Stwadd Houseware Co., Ltd
Product Name:	Electric water boiler
Brand Name:	N/A
Model No.:	DS343, DS344, DS345, DS341, DS342
Remark:	Only the model DS343 was tested, The electrical circuit design, layout, components used and internal wiring are identical, Only the model name and appearance are different.
Date of Receipt :	Sep.08,2025
Date of Test:	Sep.09,2025
Date of Report:	Sep.10,2025
Prepared by:	Shenzhen Most Technology Service Co., Ltd.

The EMC testing has been performed on the submitted samples and found in compliance with the council EMC directive 2014/30/EU.

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TEST REPORT DECLARATION

Report Number	MTEB25090126	
Applicant	Yongkang Stwadd Houseware Co., Ltd	
	Qianjin Industry #8, Gushan Town, Yongkang City, Jinhua City, Zhejiang Province	
Manufacturer	Yongkang Stwadd Houseware Co., Ltd	
	Qianjin Industry #8, Gushan Town, Yongkang City, Jinhua City, Zhejiang Province	
Product	Product Name	Electric water boiler
	Model No.	DS343
	Power Supply	220-240V~, 50-60Hz, 1500/2500W
Test Result	The EUT was found compliant with the requirement(s) of the standards.	
Standard	EN IEC 55014-1:2021, EN IEC 61000-3-2:2019+A1:2021+A2:2024, EN 61000-3-3:2013+A1:2019+A2:2021, EN IEC 55014-2:2021	

*Note

The above device has been tested by Shenzhen Most Technology Service Co., Ltd. To determine the maximum emission levels emanating from the device and the severe levels of the device can endure and its performance criterion. The test record, data evaluation & Equipment Under Test (EUT) configurations represented are contained in this test report and Shenzhen Most Technology Service Co., Ltd. Is assumed full responsibility for the accuracy and completeness of test. Also, this report shows that the EUT is technically compliant with the requirement of the above standards.

This report applies to above tested sample only. This report shall not be reproduced except in full, without written approval of Shenzhen Most Technology Service Co., Ltd., this document may be altered or revised by Shenzhen Most Technology Service Co., Ltd., personal only, and shall be noted in the revision of the document.

Prepared by	Ekaterina Zhang
	Ekaterina Zhang(Engineer)
Reviewed by	Sunny Deng
	Sunny Deng(Engineer)
Approved by	Yvette
	Yvette Zhou(Manager)

SHENZHEN MOST TECHNOLOGY SERVICE CO.,LTD. APPROVED

1. GENERAL INFORMATION

1.1.Description of Device (EUT)

Description	:	Electric water boiler
Model Number	:	DS343, DS344, DS345, DS341, DS342
Remark	:	Use DS343 for all tests

1.2.Operational Mode(s) of EUT

Order Number	:	Test Mode(s)
1	:	Running(High)
2	:	Running(Low)

1.3.Test Voltage(s) of EUT

Order Number	:	Test Voltage(s)
1	:	AC 230V/50Hz

2. DESCRIPTION OF TEST STANDARD

The intention of this publication is to establish uniform requirements for the radio disturbance level of the equipment contained in the scope, to fix limits of disturbance, to describe methods of measurement and to standardize operating conditions and interpretation of results.

The following referenced standard are indispensable for the application of this report.

Referenced Description below:

EN IEC 55014-1:2021
Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus -- Part 1: Emission.

EN IEC 61000-3-2:2019+A1:2021+A2:2024
Limits for harmonic current emissions (equipment input current <= 16 A per phase).

EN 61000-3-3:2013+A1:2019+A2:2021
Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current <= 16 A per phase and not subject to conditional connection.

EN IEC 55014-2:2021
Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus -- Part 2: Immunity - Product family standard

3. LABORATORY INFORMATION

3.1. Laboratory Name

Shenzhen Most Technology Service Co., Ltd.

3.2. Location

East A, 1/F., New Aolin Factory Buiding, Langshan Erlu, North Area, Hi-Tech Industrial Park, Nanshan District, Shenzhen, Guangdong, China

3.3. Test facility

3m Anechoic Chamber	:	Nov. 28, 2012 File on Federal Communication Commission Registration Number:490827
Shielding Room	:	Nov. 28, 2012 File on Federal Communication Commission Registration Number:490827
EMC Lab.	:	Accredited by TUV Rheinland Shenzhen Audit Report: UA 50149851 Mar. 12, 2009 Accredited by Industry Canada Registration Number: 7103A-1 Oct. 22, 2012 Accredited by TIMCO Registration Number: Q1460 March 28, 2010

3.4. Measurement Uncertainty

No.	Item	Uncertainty
1.	Uncertainty for Conducted Disturbance Test	1.25dB
2.	Uncertainty for Power Clamp Test	1.15dB
3.	Uncertainty for Radiated Disturbance Test	3.15dB

4. SUMMARY OF TEST RESULTS

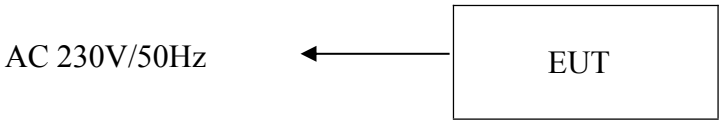
EMISSION			
Test Item	**Standard**	**Limits**	**Results**
Conducted disturbance at mains terminals	EN IEC 55014-1:2021	Class B	PASS
Radiated disturbance	EN IEC 55014-1:2021	Class B	N/A
Disturbance power test	EN IEC 55014-1:2021	---	PASS
Harmonic current emissions	EN IEC 61000-3-2:2019 +A1:2021+A2:2024	Class A	PASS
Voltage fluctuations & flicker	EN 61000-3-3:2013+A1:2019 +A2:2021	---	PASS
Clicks	EN IEC 55014-1:2021	---	PASS
IMMUNITY (EN IEC 55014-2:2021)			
Test Item	Basic Standard	Performance Criteria	Results
Electrostatic discharge (ESD)	EN 61000-4-2:2009	B	PASS
Radio-frequency, Continuous radiated disturbance	EN IEC 61000-4-3:2020	A	N/A
Electrical fast transient (EFT)	EN 61000-4-4:2012	B	PASS
Surge (Input a.c. power ports)	EN 61000-4-5:2014+A1:2017	B	PASS
Radio-frequency, Continuous conducted disturbance	EN 61000-4-6:2014	A	PASS
Voltage dips, 60% reduction	EN IEC 61000-4-11:2020	C	PASS
Voltage dips, 30% reduction	EN IEC 61000-4-11:2020	C	PASS
Voltage interruptions	EN IEC 61000-4-11:2020	C	PASS
N/A is an abbreviation for Not Applicable.			

Because the electronic control circuitry of EUT with no Oscillator frequency higher than 15MHz, and is power by mains only, According to EN IEC 55014-2 section 4, the EUT may be defined as category II. Radiation immunity (EN IEC 61000-4-3) need no test.

5. BLOCK DIAGRAM OF TEST SETUP

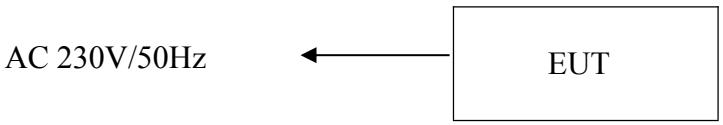
The equipments are installed test to meet EN IEC 55014-1 requirement and operating in a manner which tends to maximize its emission characteristics in a normal application. EUT was tested in normal configuration (Please See following Block diagrams)

5.1. Block Diagram of connection between EUT and simulation-EMI



(EUT: Electric water boiler)

5.2. Block Diagram of connection between EUT and simulation-EMS



(EUT: Electric water boiler)

6. TEST INSTRUMENT USED

6.1.For Conducted Disturbance at Mains Terminals Emission Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Test Receiver	Rohde & Schwarz	ESPI	101202	Mar. 04, 25	1 Year
2.	L.I.S.N.	Rohde & Schwarz	ENV216	100093	Mar. 04, 25	1 Year
3.	Coaxial Switch	Anritsu Corp	MP59B	6200283933	Mar. 04, 25	1 Year
4.	Terminator	Hubersuhner	50Ω	No.1	Mar. 04, 25	1 Year
5.	RF Cable	SchwarzBeck	N/A	No.1	Mar. 04, 25	1 Year
6.	Testing software	Fala	EZ-EMC(CE)	Ver.con-03A1	/	/

6.2.For Radiation Test (In Anechoic Chamber)(Below 1000MHz)

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Test Receiver	Rohde & Schwarz	ESCI	100492	Mar. 04, 25	1 Year
2.	Bilog Antenna	Sunol	JB3	A121206	Aug. 13, 25	1 Year
3.	Cable	Times	N/A	NO.1	Mar. 04, 25	1 Year
4.	Cable	Times	N/A	NO.2	Mar. 04, 25	1 Year
5.	Cable	Times	N/A	NO.3	Mar. 04, 25	1 Year
6.	DC Power Filter	DuoJi	DL2&30B	N/A	N/A	N/A
7.	Single Phase Power Line Filter	DuoJi	FNF 202B30	N/A	N/A	N/A
8.	3 Phase Power Line Filter	DuoJi	FNF 402B30	N/A	N/A	N/A
9.	Testing software	Fala	EZ-EMC(RE)	Ver.FA-03A1	/	/

6.3.For Radiation Test (In Anechoic Chamber)(Above 1000MHz)

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Spectrum Analyzer	Agilent	N9020A	/	Mar. 04, 25	1 Year
2.	Pre- Amplifier	MW	KA-LNA18-40-01	24001	Mar. 04, 25	1 Year
3.	Horn Antenna	Schwarzback	BBHA9120D	D69250	Mar. 16, 25	1 Year
4.	RF Cable(below1GHz)	Times	9kHz-1GHz	RF Cable No.1	Mar. 04, 25	1 Year
5.	RF Cable(above1GHz)	Times	1-18G	RF Cable No.2	Mar. 04, 25	1 Year
6.	DC Power Filter	DuoJi	DL2ᘔ30B	N/A	N/A	N/A
7.	Single Phase Power Line Filter	DuoJi	FNF 202B30	N/A	N/A	N/A
8.	3 Phase Power Line Filter	DuoJi	FNF 402B30	N/A	N/A	N/A
9.	Testing software	Fala	EZ-EMC(RE)	Ver.FA-03A1	/	/

6.4.For Electrostatic Discharge Immunity Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	ESD Tester	Zhongsheng	ESD-203AX	023K14538	Mar. 05, 25	1 Year

6.5.For Electrical Fast Transient/Burst Immunity Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	EFT tester	Prima	EFT61004TA	/	Mar. 04, 25	1 Year

6.6.For Surge Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Surge Generator	Prima	SUG61005TB	/	Mar. 04, 25	1 Year

6.7.For Injected Currents Susceptibility Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Signal Generator	ROHDE & SCHWARZ	SMB 100A	104277	Mar. 04, 25	1 Year
2.	CDN	LionCEL	CDN-M3-16	170702	NCR	NCR
3.	Power amplifier	rflight	NTWPA-00010475	14103467	Mar. 04, 25	1 Year
4.	Electromagnetic injection clamp	SCHAFFNER	KEMZ 801	15806	Mar. 04, 25	1 Year
5.	CDN	CYBERTEK	EM 5070	/	Mar. 04, 25	1 Year
6.	Attenuator	SUNWAVE	SJ-50-06DB	SW0102021339 4	Mar. 04, 25	1 Year

6.8.For Voltage Dips and Interruptions Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Cycle drop generator	Prima	DRP61011TA	/	Mar. 04, 25	1 Year

6.9.For RF Strength Susceptibility Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Signal Generator	IFR	2032	203002/100	Mar. 04, 25	1 Year
2.	Amplifier	A&R	150W1000	301584	NCR	NCR
3.	Dual Directional Coupler	A&R	DC6080	301508	Mar. 04, 25	1 Year
4.	Power Sensor	Anritsu	MA2491A	32263	Mar. 04, 25	1 Year
5.	Power Meter	R&S	NRVS	100444	Mar. 04, 25	1 Year
6.	Field Monitor	A&R	FM5004	300329	Mar. 04, 25	1 Year
7.	Field Probe	A&R	FP5000	300221	Mar. 04, 25	1 Year
8.	Log-periodic Antenna	A&R	AT1080	16512	Mar. 04, 25	1 Year
9.	RF Cable	MIYAZAKI	N/A	No.1/No.2	Mar. 04, 25	1 Year

6.10.For Power Clamp Test

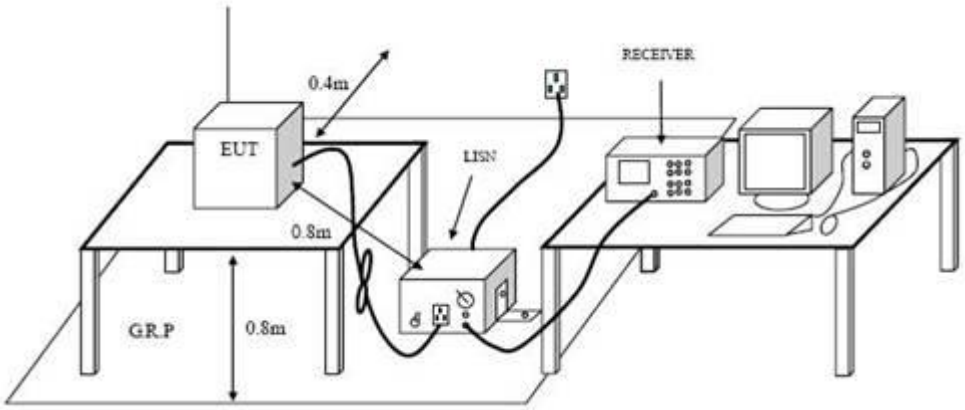
Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	Test Receiver	Rohde & Schwarz	ESCI	100492	Mar. 05, 25	1 Year
2.	Absorbing Clamp	Luthi	MDS21	3635	Mar. 05, 25	1 Year
3.	Coaxial Switch	Anritsu Corp	MP59B	6200283933	Mar. 05, 25	1 Year

6.11.For Harmonic / Flicker Test

Item	Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Cal. Interval
1.	AC Power Source	Kikusui	AC40MA	LM003232	Mar. 04, 25	1 Year
2.	Test Analyzer	Kikusui	KHA1000	LM003720	Mar. 04, 25	1 Year
3.	Line Impendence Network	Kikusui	LIN40MA-PCR-L	LM002352	Mar. 04, 25	1 Year

7. CONDUCTED DISTURBANCE AT MAINS TERMINALS TEST

7.1.Configuration of Test System



7.2.Test Standard

EN IEC 55014-1:2021

7.3.Power Line Conducted Disturbance at Mains Terminals Limit

Frequency	Maximum RF Line Voltage	
	Quasi-Peak Level Db(μV)	Average Level Db(μV)
150kHz ~ 500kHz	66 ~ 56*	59 ~ 46*
500kHz ~ 5MHz	56	46
5MHz ~ 30MHz	60	50

Notes: 1. * Decreasing linearly with logarithm of frequency.
2. The lower limit shall apply at the transition frequencies.

7.4.Test Procedure

The EUT was placed on a non-metallic table, 80cm above the ground plane. The EUT Power connected to the power mains through a line impedance stabilization network (L.I.S.N. #1). This provided a 50-ohm coupling impedance for the EUT (Please refer to the block diagram of the test setup). Both sides of power line were checked for maximum conducted interference. In order to find the maximum emission, the relative positions of equipments and all of the interface cables were changed according to EN IEC 55014-1on Conducted Disturbance at Mains Terminals test.

The bandwidth of test receiver is set at 9kHz.

The frequency range from 150kHz to 30MHz is checked. The test result are reported on Section 7.5.

7.5. Conducted Disturbance at Mains Terminals Test Results

7.5.1. Test Results: **PASS**

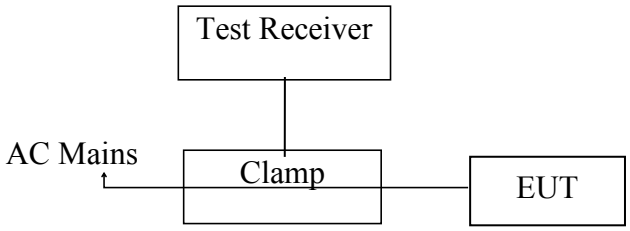
7.5.2. If the average limit is met when using a quasi-peak detector, the EUT shall be deemed to meet both limits and measurement with average detector is unnecessary.

7.5.3. Emission Level= Correct Factor + Reading Level.

7.5.4. The test data and the scanning waveform are attached within Appendix I.

8. DISTURBANCE POWER TEST

8.1. Configuration of Test System



8.2. Test Standard

EN IEC 55014-1:2021

8.3. Radiated Disturbance Limit

All emanations from devices or system including any network of conductors and apparatus connected thereto, shall not exceed the level of field strengths specified below:

FREQUENCY (MHz)	Interference Power Limits (dBpW)	
	Quasi-peak Value	Average Value
30 ~ 300	45 Increasing Linearly with Frequency to 55	35 Increasing Linearly with Frequency to 45

8.4. Test Procedure

The EUT is placed on the table which is high 0.8m by insulating support and away from other metallic surface at least 0.4m. It is connected to the power mains through an extension cord of 6m minimums. The absorber clamp was clamps the cord and moves from the far end to EUT to measure the disturbing energy emitted from the cord.

The bandwidth of the Test Receiver is set at 120kHz.

The frequency range from 30MHz to 300MHz is checked. The test result are reported on Section 8.5.

8.5. Radiated Disturbance Test Results

8.5.1.Test Results: **N/A**

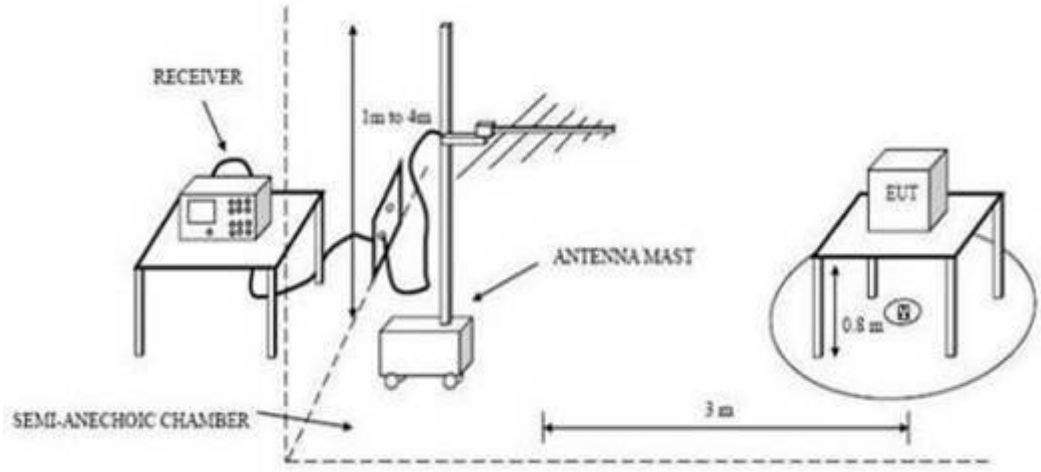
8.5.2.If the average limit is met when using a quasi-peak detector, the EUT shall be deemed to meet both limits and measurement with average detector is unnecessary.

8.5.3.Emission Level= Correct Factor + Reading Level.

8.5.4.All reading are Quasi-Peak values.

9. RADIATED DISTURBANCE TEST

9.1. Configuration of Test System



9.2. Test Standard

EN IEC 55014-1:2021

9.3. Radiated Disturbance Limit

All emanations from devices or system, including any network of conductors and apparatus connected thereto, shall not exceed the level of field strengths specified below:

FREQUENCY (MHz)	DISTANCE (Meters)	FIELD STRENGTHS LIMITS (dBμV/m)
30 ~ 230	3	40
230 ~ 1000	3	47

Note: 1.The lower limit shall apply at the transition frequencies.
2. Distance refers to the distance in meters between the test antenna and the closed point of any part of the EUT.

9.4. Test Procedure

The EUT was placed on a non-metallic table, 80 cm above the ground plane inside a semi-anechoic chamber. An antenna was located 10m from the EUT on an adjustable mast. A pre-scan was first performed in order to find prominent radiated emissions. For final emissions measurements at each frequency of interest, the EUT were rotated and the antenna height was varied between 1m and 4m in order to maximize the emission. Measurements in both horizontal and vertical polarities were made and the data was recorded. In order to find the maximum emission, the relative positions of equipments and all of the interface cables were changed according to EN IEC 55014-1 on Radiated Disturbance test.
The bandwidth setting on the test receiver is 120 kHz.
The frequency range from 30MHz to 1000MHz is checked. The test result are reported on Section 9.5.

9.5. Radiated Disturbance Test Results

9.5.1.Test Results: **PASS**

9.5.2.Emission Level= Correct Factor + Reading Level.

9.5.3.All reading are Quasi-Peak values.

9.5.4.The test data and the scanning waveform are attached within Appendix II.

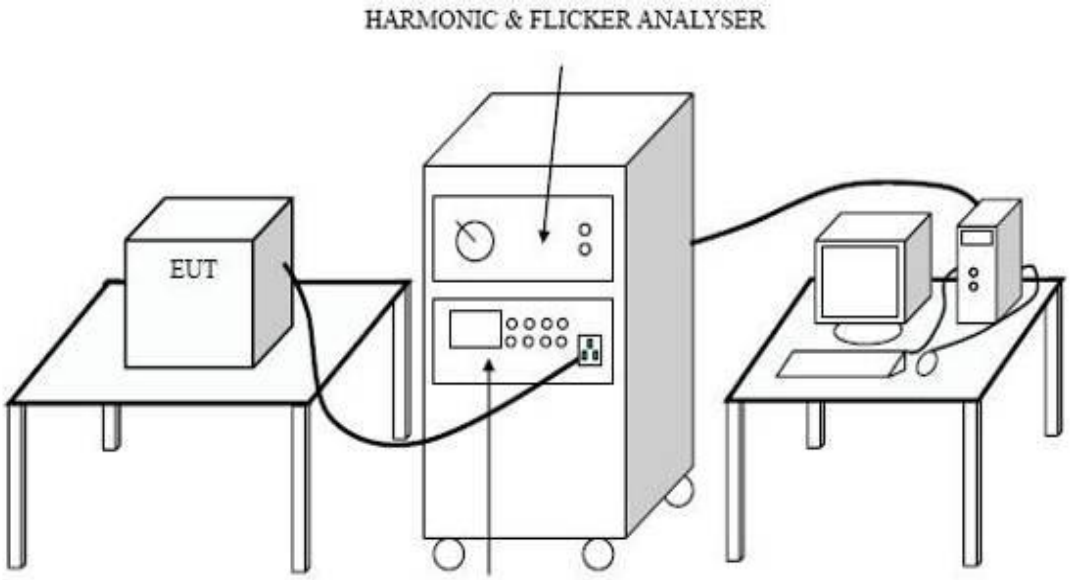
All test modes are performed, only the worst case is recorded in this report.

10. CLICKS

The EUT which fulfil the following condition:
--the click rate is no more than 5;
--none of the caused clicks has a duration longer than 20 ms,
--90% of the caused clicks have a duration less than 10 ms (measured duration time is 0.4ms), was deemed to comply with the limits.

11. HARMONIC CURRENT TEST

11.1. Configuration of Test System



11.2. Test Standard

EN IEC 61000-3-2:2019+A1:2021+A2:2024 Class A

Harmonic order h	Maximum permissible harmonic current A
Odd harmonics	
3	2,30
5	1,14
7	0,77
9	0,40
11	0,33
13	0,21
$15 \leq h \leq 39$	$0,15 \frac{15}{h}$
Even harmonics	
2	1,08
4	0,43
6	0,30
$8 \leq h \leq 40$	$0,23 \frac{8}{h}$

11.3. Test Results

11.3.1. Test Results: **PASS**

EUT:	Electric water boiler	M/N:	DS343
Mode:	Running(High)	Power:	AC 230V/50Hz
Tested by:	Peter	Test date:	2025-09-09
Temperature: / Humidity	25.8℃/ 54%		

Test Data of Current Harmonics

FINAL TEST RESULT PASS

Voltage	229.66V
Current	10.785A
Power	2476.60W
Apparent Power	2476.6VA

THC	0.008A
POHC/LIMIT	0.003A/ -----A *4
Nominal	230V/ 50Hz
Fundamental current	10.785A
Measuring period	150s
Margin	100%

HarmOrder	Limit1(A rms)	Limit2(A rms)	Ave(A rms)	Max(A rms)	LimitOver(s)	Judge
2	1.0800	2.1600	0.0025	0.003	0.0	N/A
3	2.3000	4.6000	0.0023	0.005	0.0	N/A
4	0.4300	0.8600	0.0010	0.002	0.0	N/A
5	1.1400	2.2800	0.0010	0.002	0.0	N/A
6	0.3000	0.6000	0.0010	0.001	0.0	N/A
7	0.7700	1.5400	0.0010	0.001	0.0	N/A
8	0.2300	0.4600	0.0010	0.001	0.0	N/A
9	0.4000	0.8000	0.0010	0.001	0.0	N/A
10	0.1840	0.3680	0.0010	0.001	0.0	N/A
11	0.3300	0.6600	0.0010	0.001	0.0	N/A
12	0.1533	0.3066	0.0010	0.001	0.0	N/A
13	0.2100	0.4200	0.0010	0.001	0.0	N/A
14	0.1314	0.2628	0.0010	0.001	0.0	N/A
15	0.1500	0.3000	0.0010	0.001	0.0	N/A
16	0.1150	0.2300	0.0010	0.001	0.0	N/A
17	0.1324	0.2648	0.0010	0.001	0.0	N/A
18	0.1022	0.2044	0.0010	0.001	0.0	N/A
19	0.1184	0.2368	0.0010	0.001	0.0	N/A
20	0.0920	0.1840	0.0010	0.001	0.0	N/A
21	0.1071	0.2142	0.0010	0.001	0.0	N/A
22	0.0836	0.1672	0.0010	0.001	0.0	N/A
23	0.0978	0.1956	0.0010	0.001	0.0	N/A
24	0.0767	0.1534	0.0010	0.001	0.0	N/A
25	0.0900	0.1800	0.0010	0.001	0.0	N/A
26	0.0708	0.1416	0.0010	0.001	0.0	N/A
27	0.0833	0.1666	0.0010	0.001	0.0	N/A
28	0.0657	0.1314	0.0010	0.001	0.0	N/A
29	0.0776	0.1552	0.0010	0.001	0.0	N/A
30	0.0613	0.1226	0.0010	0.001	0.0	N/A
31	0.0726	0.1452	0.0010	0.001	0.0	N/A
32	0.0575	0.1150	0.0010	0.001	0.0	N/A
33	0.0682	0.1364	0.0010	0.001	0.0	N/A
34	0.0541	0.1082	0.0010	0.001	0.0	N/A
35	0.0643	0.1286	0.0010	0.001	0.0	N/A
36	0.0511	0.1022	0.0010	0.001	0.0	N/A
37	0.0608	0.1216	0.0010	0.001	0.0	N/A
38	0.0484	0.0968	0.0010	0.001	0.0	N/A
39	0.0577	0.1154	0.0010	0.001	0.0	N/A
40	0.0460	0.0920	0.0010	0.001	0.0	N/A

Type of equipment behaviour:Quasi-Stationary

*4 When the Edition 3.0 is applied, 200% of the Limit1 value is applied to each Limit2 value.

12. VOLTAGE FLUCTUATIONS & FLICKER TEST

12.1. Configuration of Test System

Same as Section 11.1.

12.2. Test Standard

EN 61000-3-3:2013+A1:2019+A2:2021

12.3. Test Limits

The limits shall be applicable to voltage fluctuations and flicker at the supply terminals of the equipment under test,The following limits apply:
the value of P_{st} shall not be greater than 1,0;
the value of P_{lt} shall not be greater than 0,65;
Tmax, the accumulated time value of d(t) with a deviation exceeding 3,3 % during a single voltage change at the EUT terminals, shall not exceed 500 ms;
the maximum relative steady-state voltage change, dc, shall not exceed 3,3 %;
the maximum relative voltage change dmax, shall not exceed:
a) 4% without additional conditions;
b) 6 % for equipment which is:
switched manually, or switched automatically more frequently than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds), or manual restart, after a power supply interruption.
c) 7 % for equipment which is:
attended whilst in use (for example: hair dryers, vacuum cleaners, kitchen equipment such as mixers, garden equipment such as lawn mowers, portable tools such as electric drills), or
switched on automatically, or is intended to be switched on manually, no more than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds) or manual restart, after a power supply interruption.

12.4. Test Results

12.4.1. Test Results: **PASS**

12.4.2. All test data on the following pages.

EUT:	**Electric water boiler**	**M/N:**	**DS343**
Mode:	**Running(High)**	**Power:**	**AC 230V/50Hz**
Tested by:	**Peter**	**Test date:**	**2025-09-09**
Temperature: / Humidity	**25.8℃/ 54%**		

Test Data of Voltage Fluction and Flicker

FINAL TEST RESULT	PASS
Nominal Voltage	230V
Nominal Frequency	50Hz
Plt Test duration	630s
Flicker Margin	100%
d Measurement Margin	100%

	Pst	dc(%)	dmax(%)	d(t)>3.3%(ms)	Judge
Limit	1.000	3.300	4.000	500	
Seg. 1	0.201	0.004	0.030	0	Pass
Seg. 2	-.---	-.---	-.---	-----	
Seg. 3	-.---	-.---	-.---	-----	
Seg. 4	-.---	-.---	-.---	-----	
Seg. 5	-.---	-.---	-.---	-----	
Seg. 6	-.---	-.---	-.---	-----	
Seg. 7	-.---	-.---	-.---	-----	
Seg. 8	-.---	-.---	-.---	-----	
Seg. 9	-.---	-.---	-.---	-----	
Seg.10	-.---	-.---	-.---	-----	
Seg.11	-.---	-.---	-.---	-----	
Seg.12	-.---	-.---	-.---	-----	

	Plt	Judge
Limit	0.650	
Measurement	0.088	Pass

13. IMMUNITY PERFORMANCE CRITERIA

The test results shall be classified in terms of the loss of function or degradation of performance of the equipment under test, relative to a performance level by its manufacturer or the requestor of the test, or the agreed between the manufacturer and the purchaser of the product.

Definition related to the performance level:
Based on the used product standard
Based on the declaration of the manufacturer, requestor or purchaser

Criterion A:
The apparatus shall continue to operate as intended during the test and after the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

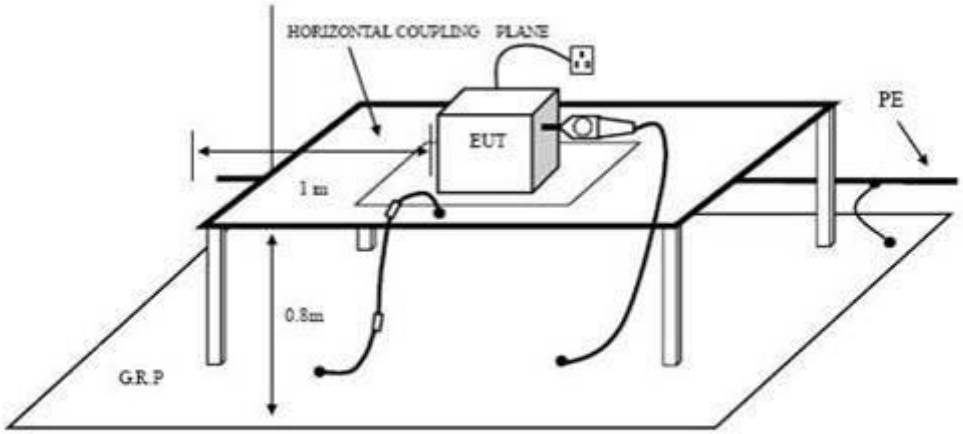
Criterion B:
The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed, however. No change of actual operation state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonably expect form the apparatus the apparatus if used as intended.

Criterion C:
Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or by any operation specified in the instructions for use.

14. ELECTROSTATIC DISCHARGE IMMUNITY TEST

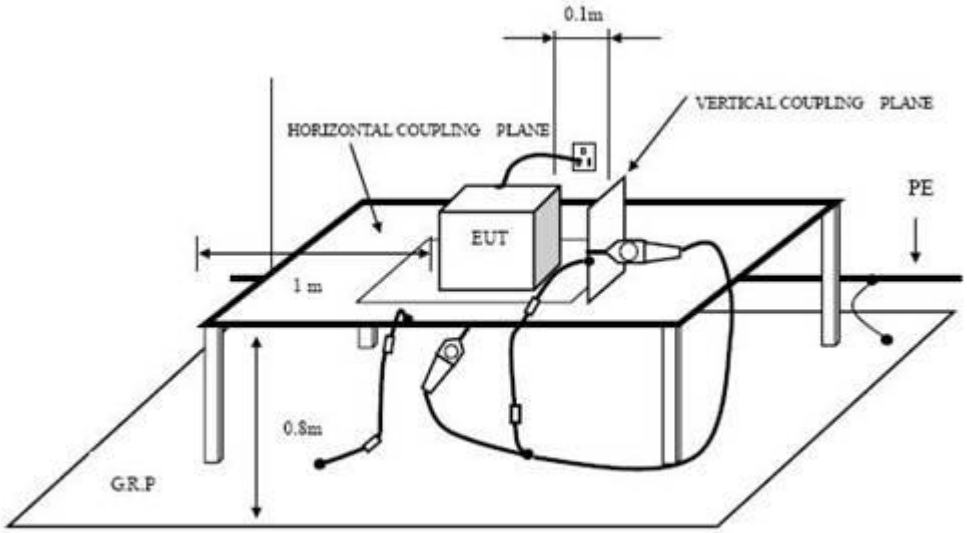
14.1. Configuration of Test System

14.1.1. Configuration of ESD Test System(Direct Discharge)



DIRECT DISCHARGE SETUP

14.1.2. Configuration of ESD Test System(Indirect Discharge)



INDIRECT DISCHARGE SETUP

14.2. Test Standard

EN IEC 55014-2:2021(EN 61000-4-2)
(Severity Level 3 for Air Discharge at 8KV,
Severity Level 2 for Contact Discharge at 4KV)

14.3.Severity Levels and Performance Criterion

14.3.1.Severity level

Level	Test Voltage Contact Discharge (KV)	Test Voltage Air Discharge (KV)
1.	2	2
2.	4	4
3.	6	8
4.	8	15
X	Special	Special

14.3.2.Performance criterion : **B**

14.4.Test Procedure

14.4.1.Air Discharge**:**

The test was applied on non-conductive surfaces of EUT. The round discharge tip of the discharge electrode was approached as fast as possible to touch the EUT. After each discharge, the discharge electrode was removed from the EUT. The generator was re-triggered for a new single discharge and repeated 20 times for each pre-selected test point. This procedure was repeated until all the air discharge completed

14.4.2.Contact Discharge:

All the procedure was same as Section 14.4.1. except that the generator was re-triggered for a new single discharge for each pre-selected test point. The tip of the discharge electrode was touch the EUT before the discharge switch was operated.

14.4.3.Indirect discharge for horizontal coupling plane

At least 20 single discharges were applied to the horizontal coupling plane, at points on each side of the EUT. The discharge electrode positions vertically at a distance of 0.1m from the EUT and with the discharge electrode touching the coupling plane.

14.4.4.Indirect discharge for vertical coupling plane

At least 20 single discharge were applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, was placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges were applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

14.5.Test Results

14.5.1.Test Results: **PASS**

14.5.2.Test data on the following pages.

Electrostatic Discharge Test Results

Shenzhen Most Technology Service Co., Ltd.

Test Voltage :	*1*	*Test Date:*	*Sep.09,2025*
Test Mode :	*1&2*	*Criterion :*	*B*
Temperature:	*25.8℃*	*Humidity:*	*54%*

Air Discharge: ±8KV *# For Air Discharge each Point Positive 10 times and negative 10 times discharge.*

Contact Discharge: ±4KV *# For Contact Discharge each point positive 10 times and negative 10 times discharge*

Test Results Description

Location	***Kind*** *A-Air Discharge C-Contact Discharge*	***Result***
Gaps	*A*	*PASS*
Keys	*A*	*PASS*
Screws	*C*	*PASS*
HCP	*C*	*PASS*
VCP of Front	*C*	*PASS*
VCP of Rear	*C*	*PASS*
VCP of Left	*C*	*PASS*
VCP of Right	*C*	*PASS*

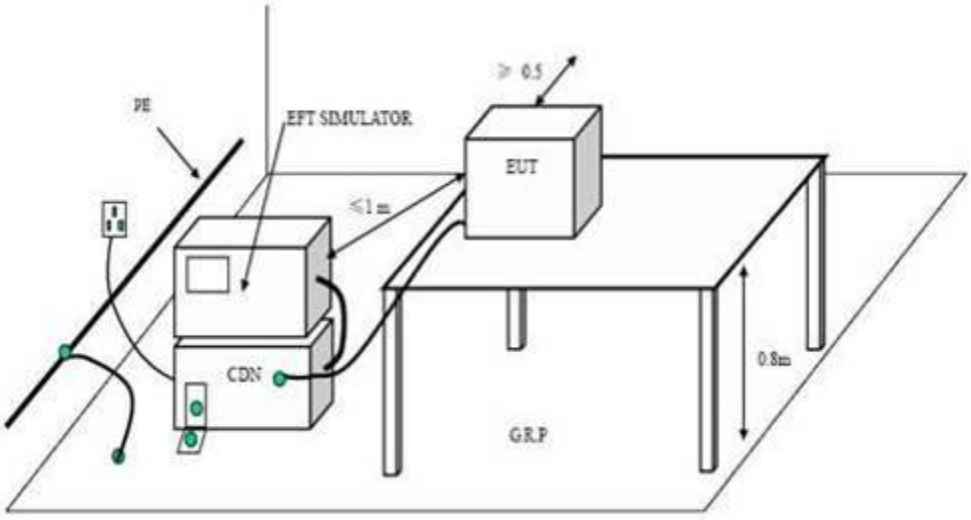
Remark :No obvious abnormal phenomena.

Discharge was considered on Contact and Air and Horizontal Coupling Plane (HCP) and Vertical Coupling Plane (VCP).

Reviewer : Sunny

15. ELECTRICAL FAST TRANSIENT/BURST IMMUNITY TEST

15.1. Configuration of Test System



15.2. Test Standard

EN IEC 55014-2:2021 (EN 61000-4-4)
(Severity Level 2 at 1KV)

15.3. Severity Levels and Performance Criterion

15.3.1. Severity level

Open Circuit Output Test Voltage ±10%		
Level	On Power Supply Lines	On I/O (Input/Output) Signal lines and control lines
1.	0.5 KV	0.25 KV
2.	1 KV	0.5 KV
3.	2 KV	1 KV
4.	4 KV	2 KV
X	Special	Special

15.3.2. Performance criterion : **B**

15.4. Test Procedure

The EUT and its simulators were placed on a the ground reference plane and were insulated from it by an wood support 0.1m ± 0.01m thick. The ground reference plane was 1m*1m metallic sheet with 0.65mm minimum thickness. This reference ground plane was project beyond the EUT by at least 0.1m on all sides and the minimum distance between EUT and all other conductive structure, except the ground plane was more than 0.5m. All cables to the EUT was placed on the wood support, cables not subject to EFT/B was routed as far as possible from the cable under test to minimize the coupling between the cables.

15.4.1. For input and AC power ports**:**

The EUT was connected to the power mains by using a coupling device which couples the EFT interference signal to AC power lines. Both positive transients and negative transients of test voltage was applied during compliance test and the duration of the test can't less than 2mins.

15.4.2. For signal lines and control lines ports:

It's not I/O ports,
It's unnecessary to test.

15.4.3. For DC input and DC output power ports:

It's not I/O ports,
It's unnecessary to test.

15.5. Test Results

15.5.1. Test Results: **PASS**

15.5.2. Test data on the following pages.

Electrical Fast Transient/Burst Test Results

Shenzhen Most Technology Service Co., Ltd.

Test Voltage :	1	Test Date:	Sep.09,2025
Test Mode :	1&2	Criterion :	B
Temperature:	25.8℃	Humidity:	54%

Test Results Description

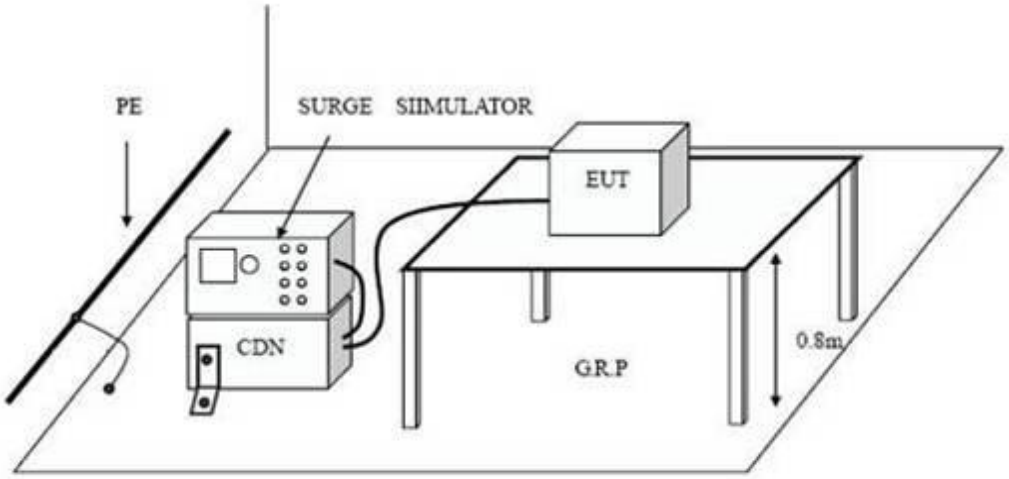
Inject Line	Voltage KV	Inject Time(s)	Inject Method	Results	Inject Line	Voltage KV	Inject Time(s)	Inject Method	Results
L	±1	120	Direct	PASS					
N	±1	120	Direct	PASS					
L+ N	±1	120	Direct	PASS					
PE	±1	120	Direct	PASS					
L+ PE	±1	120	Direct	PASS					
N+ PE	±1	120	Direct	PASS					
L+N+ PE	±1	120	Direct	PASS					

Remark: No obvious abnormal phenomena.

Reviewer : Sunny

16. SURGE TEST

16.1. Configuration of Test System



16.2. Test Standard

EN IEC 55014-2:2021 (EN 61000-4-5)
(Severity Level **:** Line to Line was Level 2 at 1KV
Line to PE was Level 3 at 2KV)

16.3. Severity Levels and Performance Criterion

16.3.1. Severity level

Severity Level	Open-Circuit Test Voltage KV
1	0.5
2	1.0
3	2.0
4	4.0
*	Special

16.3.2. Performance criterion : **B**

16.4. Test Procedure

16.4.1. Set up the EUT and test generator as shown on Section 16.1.

16.4.2. For line to line coupling mode, provide a 1KV 1.2/50us voltage surge (at open-circuit condition) and 8/20us current surge to EUT selected points, and for active line / neutral line to ground are same except test level is 2KV.

16.4.3. At least 5 positive and 5 negative (polarity) tests with a maximum 1/min repetition rate are applied during test.

16.4.4. Different phase angles are done individually.

16.4.5. Record the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.

16.5. Test Results

16.5.1. Test Results: **PASS**

16.5.2. Test data on the following pages.

Surge Immunity Test Results

Shenzhen Most Technology Service Co., Ltd.

Test Voltage :	1	Test Date:	Sep.09,2025
Test Mode :	1&2	Criterion	B
Temperature:	25.8℃	Humidity:	54%

Test Results Description

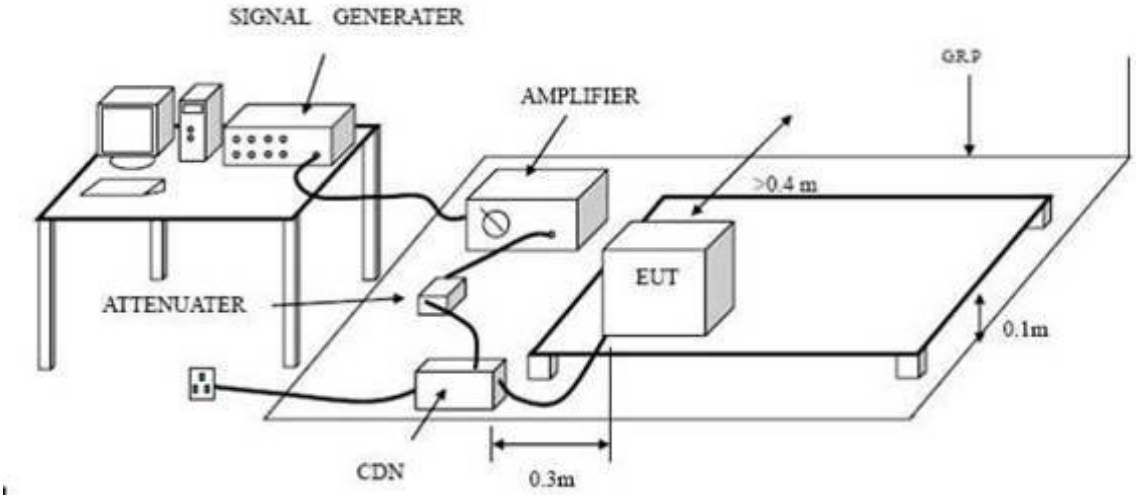
Location	Polarity	Phase Angle	No of Pulse	Pulse Voltage (KV)	Result
L-N	+	90	5	1	PASS
	-	270	5	1	PASS
L-PE	+	90	5	2.0	PASS
	-	270	5	2.0	PASS
N- PE	+	90	5	2.0	PASS
	-	270	5	2.0	PASS
L+ N- PE	+	90	5	2.0	PASS
	-	270	5	2.0	PASS

Remark:No obvious abnormal phenomena.

Reviewer : Sunny

17. INJECTED CURRENTS SUSCEPTIBILITY TEST

17.1.Configuration of Test System



17.2.Test Standard

EN IEC 55014-2:2021 (EN 61000-4-6)
(Severity Level 2 at 3V (r.m.s.) and frequency is from 0.15MHz to 230MHz)

17.3. Severity Levels and Performance Criterion

17.3.1.Severity level

Level	Voltage Level (e.m.f.) V
1.	1
2.	3
3.	10
X	Special

17.3.2.Performance criterion: **A**

17.4. Test Procedure

17.4.1. Set up the EUT, CDN and test generators as shown on Section 17.1.

17.4.2. Let the EUT work in test mode and test it.

17.4.3. The EUT are placed on an insulating support 0.1m high above a ground reference plane. CDN (coupling and decoupling device) is placed on the ground plane about 0.3m from EUT. Cables between CDN and EUT are as short as possible, and their height above the ground reference plane shall be between 30 and 50 mm (where possible).

17.4.4. The disturbance signal described below is injected to EUT through CDN.

17.4.5. The EUT operates within its operational mode(s) under intended climatic conditions after power on.

17.4.6. The frequency range is swept from 0.150MHz to 230MHz using 3V signal level, and with the disturbance signal 80% amplitude modulated with a 1KHz sine wave.

17.4.7. The rate of sweep shall not exceed 1.5*10-3decades/s. Where the frequency is swept incrementally, the step size shall not exceed 1% of the start and thereafter 1% of the preceding frequency value.

17.4.8. Recording the EUT operating situation during compliance testing and decide the EUT immunity criterion.

17.5. Test Results

17.5.1. Test Results: **PASS**

17.5.2. Test data on the following pages.

Injected Currents Susceptibility Test Results

Shenzhen Most Technology Service Co., Ltd.

Test Voltage :	*1*	*Test Date:*	*Sep.09,2025*
Test Mode :	*1&2*	*Criterion :*	*A*
Temperature:	*25.8 ℃*	*Humidity:*	*54%*

Test Results Description

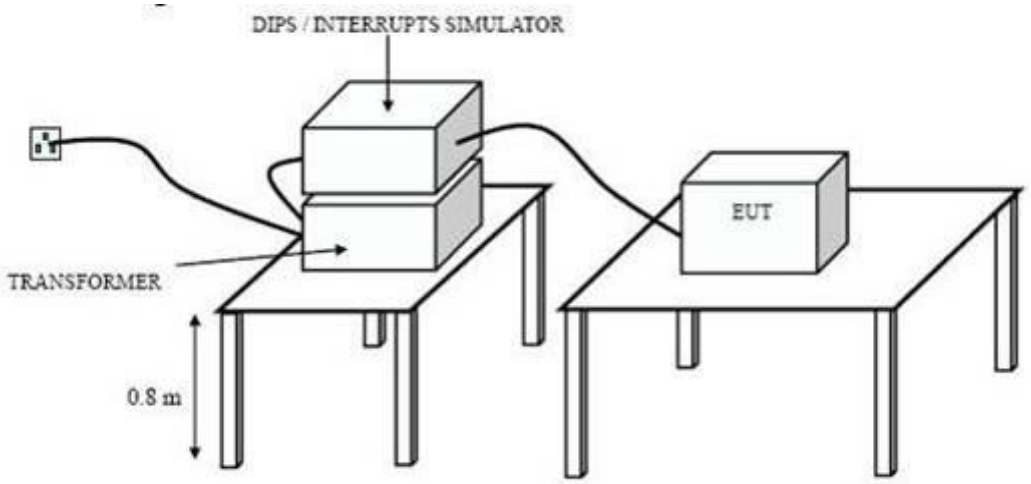
Frequency Range (MHz)	*Injected Position*	*Voltage Level (e.m.f.)*	*Criterion*	*Result*
0.15 ~ 230	*AC Mains*	*3V(rms), Unmodulated*	*A*	*PASS*

Remark : No obvious abnormal phenomena.

Reviewer : Sunny

18. VOLTAGE DIPS AND INTERRUPTIONS TEST

18.1. Configuration of Test System



18.2. Test Standard

EN IEC 55014-2:2021(EN IEC 61000-4-11)
(Severity level: 0% 0.5 period
70% 25 periods
40% 10 periods)

18.3. Severity Levels and Performance Criterion

18.3.1. Severity level

Test Level %U_T	Voltage dip and short interruptions %U_T	Performance Criterion	Duration (in period)
0	100	C	0.5
40	60	C	10
70	30	C	25

18.3.2. Performance criterion **: C**

18.4. Test Procedure

18.4.1. Set up the EUT and test generator as shown on section 18.1.

18.4.2. The interruptions is introduced at selected phase angles with specified duration there is a 3mins minimum interval between each test event.

18.4.3. After each test a full functional check is performed before the next test.

18.4.4. Repeat procedures 2 & 3 for voltage dips only the test level and duration is changed.

18.4.5. Record any degradation of performance.

18.5. Test Results

18.5.1. Test Results: **PASS**

18.5.2. Test data on the following pages.

Voltage Dips And Interruptions Test Results

Shenzhen Most Technology Service Co., Ltd.

Test Voltage :	1	Test Date:	Sep.09,2025
Test Mode :	1&2	Criterion	C
Temperature:	25.8 ℃	Humidity:	54%

Test Results Description

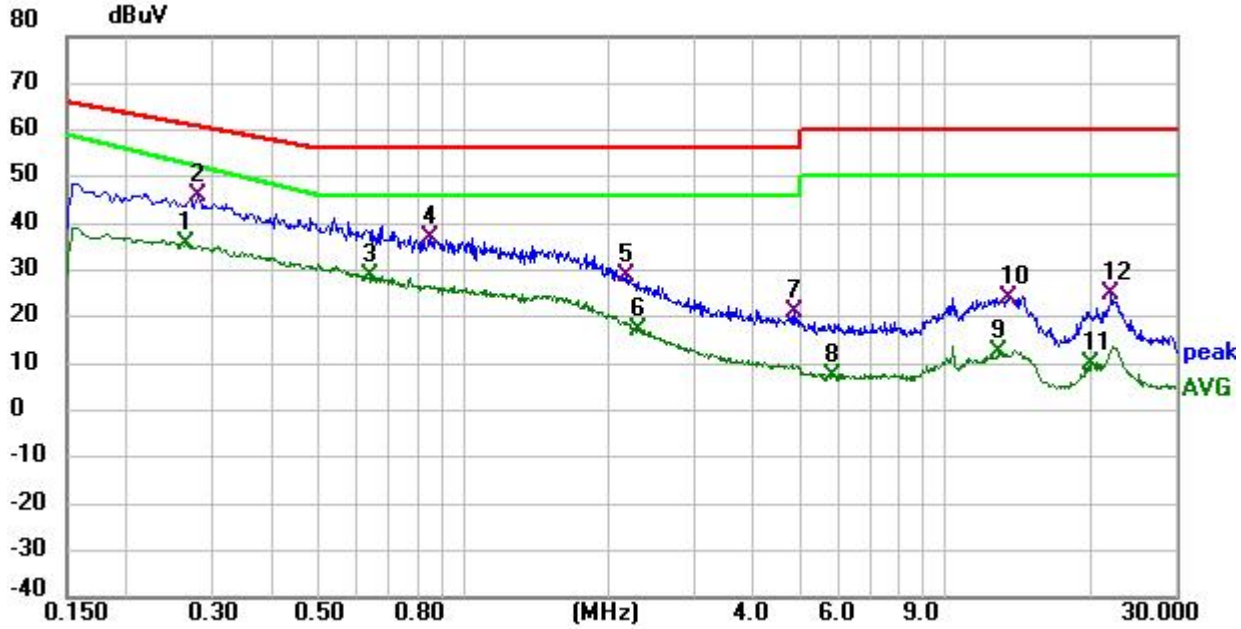
Test Level % U_T	Voltage Dips & Short Interruptions % U_T	Duration (in period)	Phase Angle	Criterion	Result
0	100	0.5P	0°~360°	C	PASS
70	30	25P	0°~360°	C	PASS
40	60	10P	0°~360°	C	PASS

Remark: 1: UT is the rated voltage for the equipment.

Reviewer : Sunny

APPENDIX I

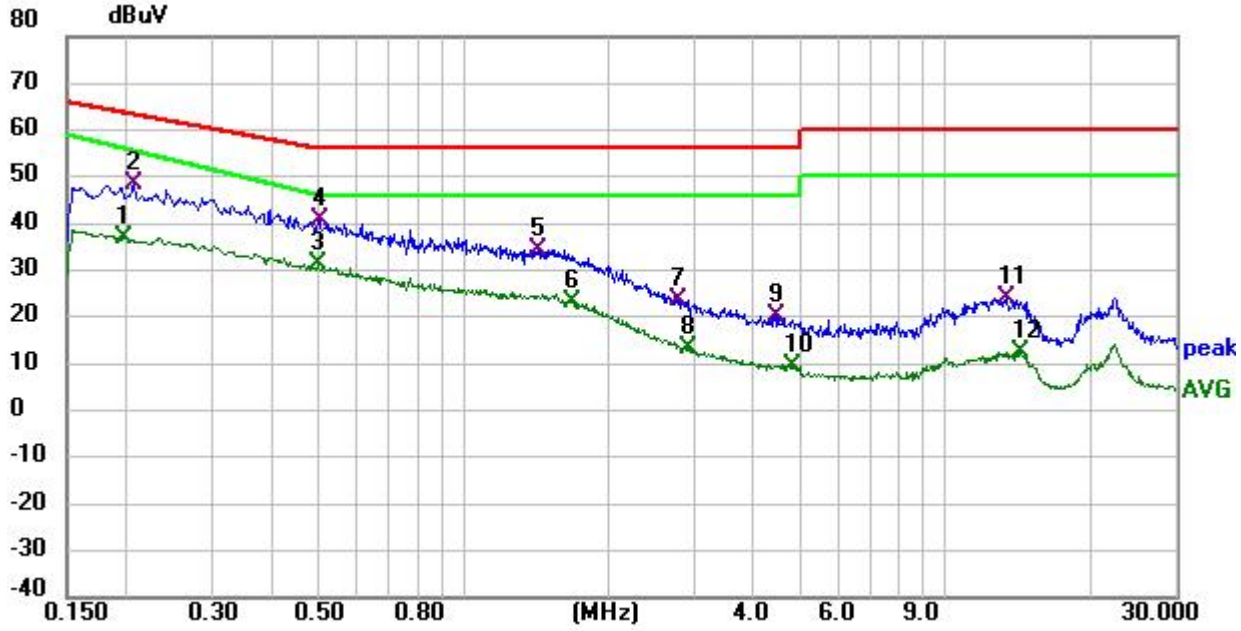
EUT:	Electric water boiler	M/N:	DS343
Mode:	Running(High)	Phase:	L1
Tested by:	Cole	Power:	AC 230V/50Hz
Temperature: / Humidity	25.8℃/ 54%	Test date:	2025-09-09



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1	0.2660	26.14	9.60	35.74	52.81	-17.07	AVG	P	
2 *	0.2819	36.34	9.59	45.93	60.76	-14.83	QP	P	
3	0.6380	19.36	9.59	28.95	46.00	-17.05	AVG	P	
4	0.8500	27.39	9.60	36.99	56.00	-19.01	QP	P	
5	2.1740	19.24	9.60	28.84	56.00	-27.16	QP	P	
6	2.3060	7.57	9.61	17.18	46.00	-28.82	AVG	P	
7	4.8940	11.52	9.63	21.15	56.00	-34.85	QP	P	
8	5.8300	-2.10	9.64	7.54	50.00	-42.46	AVG	P	
9	12.8900	2.88	9.70	12.58	50.00	-37.42	AVG	P	
10	13.5740	14.54	9.70	24.24	60.00	-35.76	QP	P	
11	20.0900	0.12	9.73	9.85	50.00	-40.15	AVG	P	
12	22.0860	15.03	9.74	24.77	60.00	-35.23	QP	P	

*:Maximum data x:Over limit !:over margin

EUT:	Electric water boiler	M/N:	DS343
Mode:	Running(High)	Phase:	N
Tested by:	Cole	Power:	AC 230V/50Hz
Temperature: / Humidity	25.8℃/ 54%	Test date:	2025-09-09

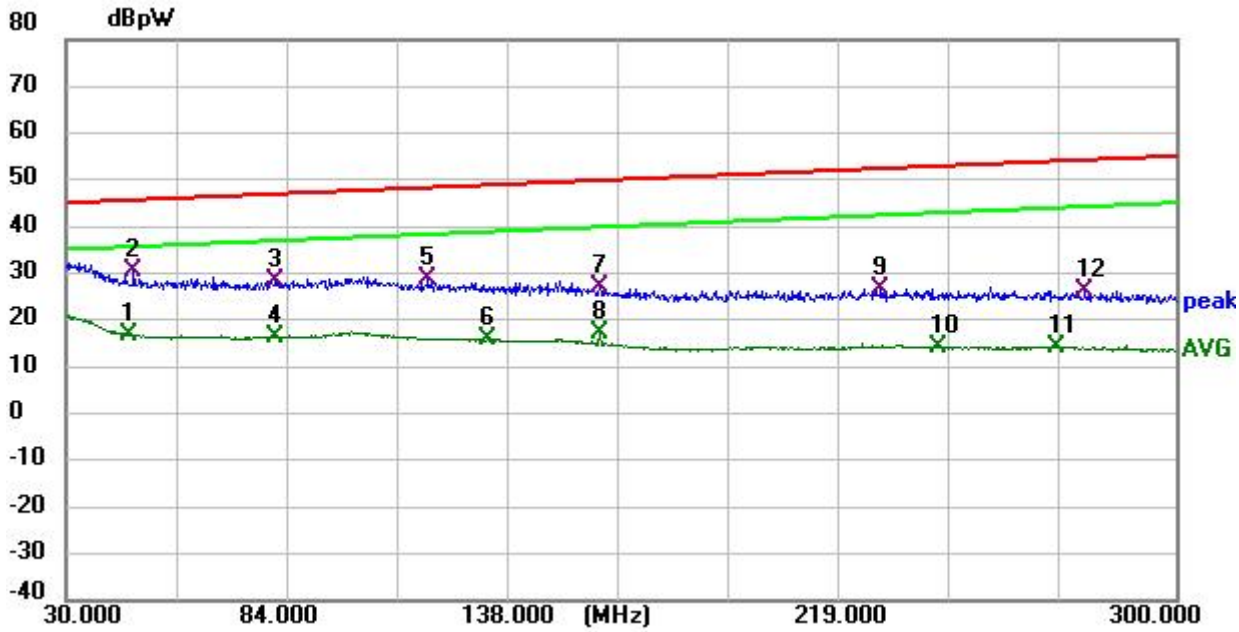


No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1	0.1980	27.47	9.60	37.07	56.00	-18.93	AVG	P	
2	0.2060	38.98	9.60	48.58	63.37	-14.79	QP	P	
3 *	0.4980	21.76	9.59	31.35	46.04	-14.69	AVG	P	
4	0.5020	31.34	9.59	40.93	56.00	-15.07	QP	P	
5	1.4299	24.86	9.60	34.46	56.00	-21.54	QP	P	
6	1.6820	13.55	9.60	23.15	46.00	-22.85	AVG	P	
7	2.7940	14.18	9.61	23.79	56.00	-32.21	QP	P	
8	2.9180	3.75	9.61	13.36	46.00	-32.64	AVG	P	
9	4.4580	10.71	9.62	20.33	56.00	-35.67	QP	P	
10	4.8020	-0.24	9.63	9.39	46.00	-36.61	AVG	P	
11	13.4180	14.37	9.70	24.07	60.00	-35.93	QP	P	
12	14.3060	2.73	9.70	12.43	50.00	-37.57	AVG	P	

*:Maximum data x:Over limit !:over margin

APPENDIX II

EUT:	Electric water boiler	M/N:	DS343
Mode:	Running(High)	Note:	DC Line
Tested by:	Cole	Power:	AC 230V/50Hz
Temperature: / Humidity	25.8℃/ 54%	Test date:	2025-09-09



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBpW)	Limit (dBpW)	Margin (dB)	Detector	Position (cm)	P/F	Remark
1	45.6000	-0.75	17.67	16.92	35.58	-18.66	AVG		P	
2 *	46.5200	12.91	17.55	30.46	45.61	-15.15	QP		P	
3	81.2800	11.37	17.11	28.48	46.90	-18.42	QP		P	
4	81.2800	-0.58	17.11	16.53	36.90	-20.37	AVG		P	
5	118.2400	12.13	16.71	28.84	48.27	-19.43	QP		P	
6	132.6000	-0.22	16.35	16.13	38.80	-22.67	AVG		P	
7	160.0000	11.65	15.60	27.25	49.81	-22.56	QP		P	
8	160.0000	1.72	15.60	17.32	39.81	-22.49	AVG		P	
9	228.2400	11.29	15.26	26.55	52.34	-25.79	QP		P	
10	242.2800	-0.75	15.05	14.30	42.86	-28.56	AVG		P	
11	270.8800	-0.66	14.97	14.31	43.92	-29.61	AVG		P	
12	277.9200	11.35	14.76	26.11	54.18	-28.07	QP		P	

*:Maximum data x:Over limit I:over margin

APPENDIX III
(Test Photos)

Conducted Test Setup Photograph



Disturbance Power Test Setup Photograph



ESD Test Setup Photograph



APPENDIX IV
(Photos of the EUT)

GENERAL VIEW OF SAMPLE



30L:

GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



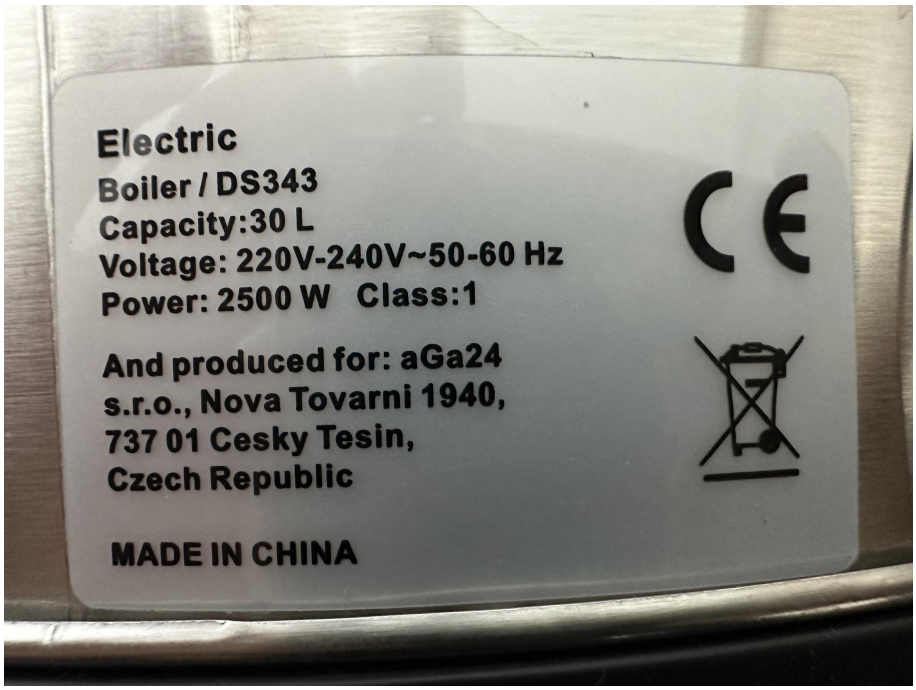
GENERAL VIEW OF SAMPLE



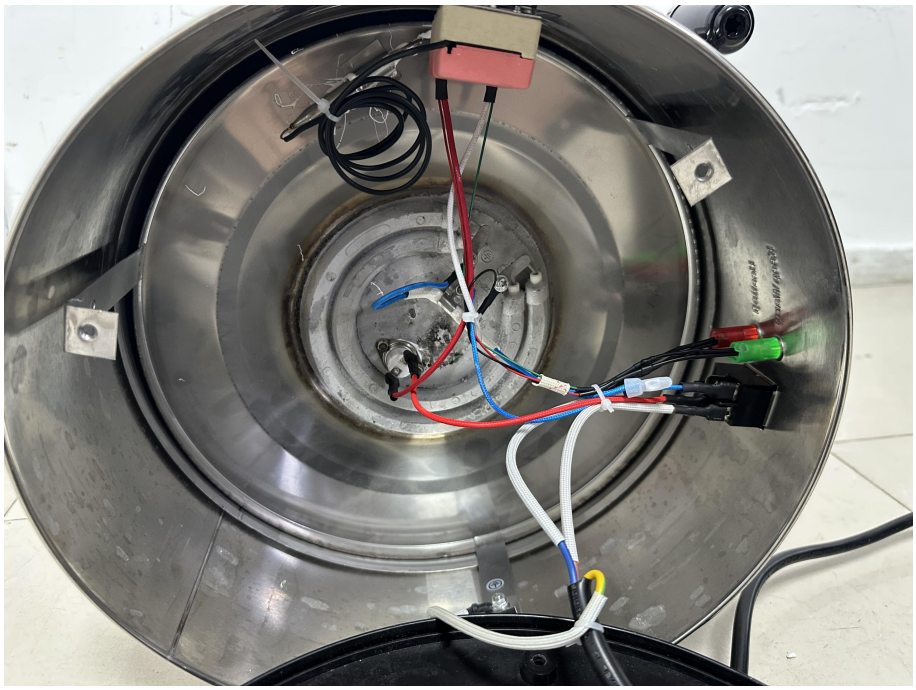
GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



20L:

GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



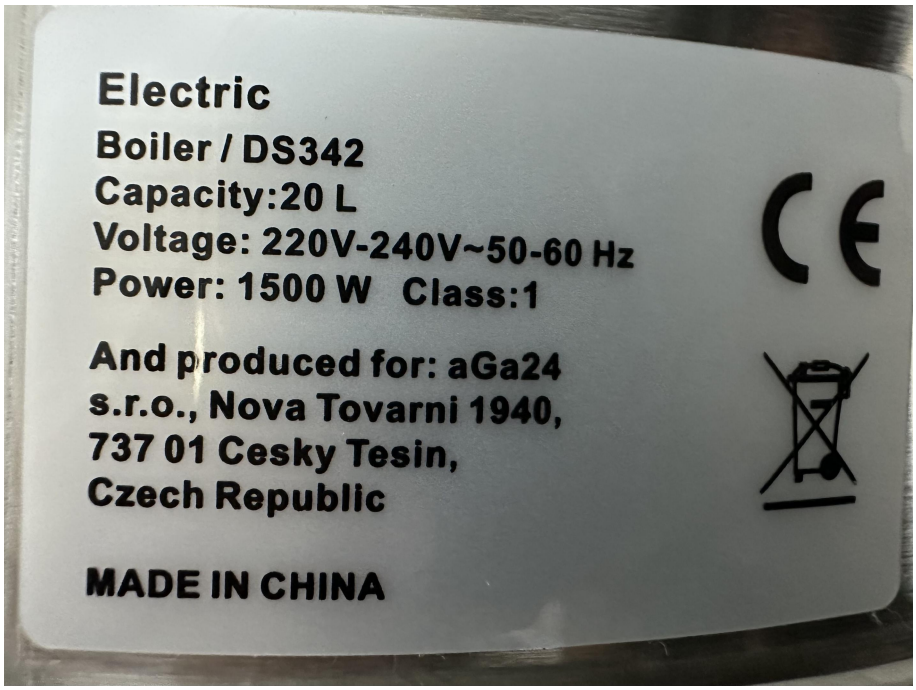
GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE



GENERAL VIEW OF SAMPLE

